

End-to-End Optical Music Recognition for Jazz Lead Sheets

Grau en Enginyeria Informàtica (Computació) · TFG

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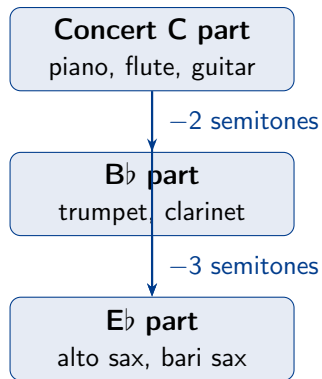
A problem every jazz musician knows

Years of jazz and big band experience. . .

- Rehearsals start, you get a stack of paper
- Brass section mixes concert-pitch and transposing instruments
 - ▶ **B $\flat$** — trumpet, clarinet, soprano sax
 - ▶ **E $\flat$** — alto sax, baritone sax
 - ▶ **F** — French horn
- You receive the lead sheet in one transposition, need it in another
- Manual on-paper transposition: slow, error-prone, and must be repeated every time

The sheet music on this table is a real example.

If it existed as structured data, transposition would be a single click.



Today: done by hand on paper.
Goal: done automatically.

The Real Book and the digitisation gap

- Most widely circulated jazz fake book in the world
- Assembled by Berklee students in the 1970s; now officially published
- Hundreds of jazz standards — on every musician's stand worldwide
- Official PDF is **image-only**: not searchable, not editable, not machine-readable



Today: scan → image PDF
(no symbolic content)

Goal: scan → structured JSON
(pitch, rhythm, chords)

What full digitisation would unlock:

- **Automatic transposition** for all instrument families
- MIDI playback for practice
- Corpus-level harmonic analysis (jazz theory research)

Four compounding challenges

1. Notation complexity

Pitch = vertical position *relative to the clef*.
Duration = head shape + stem + flags + dots. Beams create joint dependencies across groups of notes. No standalone symbols.

2. Domain gap

Models trained on clean computer-rendered images fail on real scans: noise, blur, perspective distortion, ink bleed, paper yellowing, JPEG artefacts.

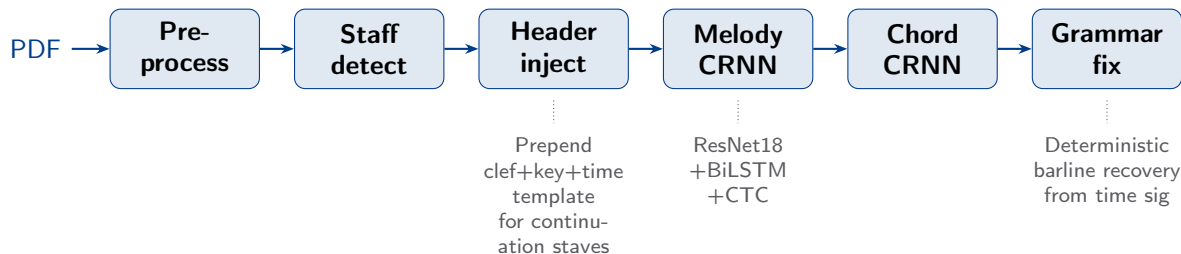
3. Data scarcity

No publicly labelled corpus of Real Book pages.
Must construct synthetic training data from scratch
in the target engraving style.

4. Dual-stream recognition

A lead sheet carries two simultaneous streams: **melody** on the staff (pattern recognition) and **chord symbols** above it (character recognition). Existing OMR systems address only the first.

Five-stage inference pipeline



Header injection (key design):

Continuation staves have no clef, key, or time signature. Injection keeps every staff in-distribution for the CRNN — zero training cost, no leakage risk.

Two parallel CRNNs:

Separate model for each visual language. Melody: LMX token sequence. Chords: character-level recognition.

End-to-end latency:

114 ms per page
(8-staff page, RTX 3060, greedy decoding)

Why CRNN+CTC?

Alternative

Modular pipeline

Seq2seq + attention

Transformer (SMT, LEGATO)

Reason rejected

Needs per-symbol XML tree. Requires full hierarchical context to emit each node. Incompatible with CTC's atomic, left-to-right token stream.

Adds encoder-decoder overhead with no benefit for left-to-right phonic reading

State-of-the-art > 10× compute single-RTX-3060

Why LMX?

✗ MusicXML

✓ LMX (chosen)

Flat, whitespace-delimited tokens.

Each token is atomic: pitch:C, quarter, flat, measure.

≈100-token vocabulary.

Regular grammar ⇒ deterministic post-pass fixer.

Backbone: ResNet18 trains in half the time of VGG at comparable SER.

Synthetic data pipeline and quality control

Building the training corpus:

- PrlMuS: 87,678 monophonic incipits
- Re-rendered in **LilyJAZZ** style (matching Real Book typography)
- 3 domain filters $\Rightarrow$ 46,089 survive
- Split: 36,873 / 4,608 / 4,608 (train/val/test)
- Each sample: 1 clean + 2 scan-simulated PNGs
- $\approx$ 166,000 rendered assets on disk

Scan simulation pipeline:

- Gaussian + salt-and-pepper noise
- Motion blur and defocus
- Perspective distortion
- Ink bleed
- Paper yellowing (tone remap)
- JPEG compression

Acceptance criterion: clean-to-scanned SER gap < 1 pp.

Achieved: $\Delta = 0.06$ pp.



The twin leakage bug — quality by finding the problem

What happened:

- ① `extra_data_dirs` added derived image copies of training samples
- ② They joined the *shared* split pool **before** the random permutation
- ③ Near-duplicate training samples appeared in the val/test splits
- ④ All metrics before 3 June 2026 were inflated

Fix applied:

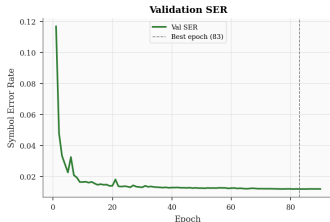
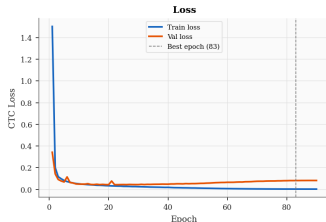
- Introduced `scanned_variant_dirs` — train-only by construction, keyed by sample ID
- Clean split protocol re-run; all results in this presentation are post-fix

Why this is a positive outcome

Identifying data contamination mid-project, designing a leakage-safe mechanism, and re-reporting clean numbers is precisely the engineering rigour the project required.

The metrics now reflect the true generalisation of the system.

Training trajectory



Observations:

- SER falls from $\approx 12\%$ to $\approx 2\%$ by epoch 7 — the alignment is learnt early
- Convergence to $\approx 1.2\%$ by epoch 83
- Validation plateau: ± 0.02 pp over last 10 epochs
- 90 epochs total, ≈ 13 h on RTX 3060

The training distribution is **scan-augmented only** — clean images are held out for evaluation, so the gap is measured fairly.

Headline results

Metric	Scanned (%)	Clean (%)
Aggregate SER	1.23	1.17
Melodic SER	0.14	0.10
Perfect transcriptions	72.7	73.7
SER $\leq$ 5%	90.3	90.7

✓ Target met

Scanned SER 1.23% < 3%
(target set at project start)

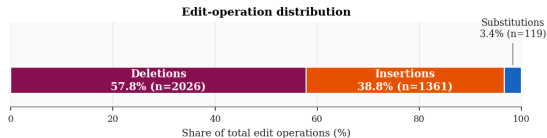
✓ Melody is accurate

99.86% of pitch, duration,
and accidental tokens are
correct

✓ Gap is negligible

Clean vs scanned: 0.06 pp
(criterion: < 1 pp)

Error analysis — where the residual error lives



3,506 total edits on the scanned split:

- $\approx 90\%$ are measure (barline) and tie tokens
- Pitch and octave tokens barely appear in deletions or insertions
- Substitutions: only 3.4% of all edits

Most common substitution:

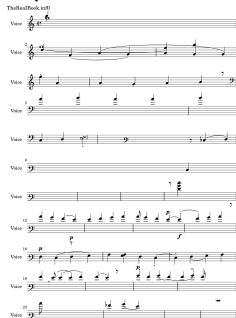
pitch:B $\rightarrow$ pitch:G (8 times over 4,608 staves)

Key takeaway

The model reads the **music** correctly. It occasionally misplaces a **barline**. The melodic SER (0.14%) measures this directly:

Audiveris 5.10 (state-of-the-art

Thumbnail.indd



open-source)

- ✗ Correct clef on **2 of 9 staves**; rest default to bass clef
- ✗ **No chord symbols** recovered
- ✗ Octaves displaced; output musically unusable



This system

- ✓ **7 of 9 staves** recognised, all on correct treble clef
- ✓ **36 chord symbols** recovered:
D-7 G7 / E-7 A7 / G-7 C7 Fmaj7 ...
- ✓ Title band + credit line correctly

Methodology: incremental — one pipeline stage at a time.

Budget: 540 h ($18 \text{ ECTS} \times 30 \text{ h/ECTS}$). On target.

Main deviation from plan:

- Twin leakage discovery mid-project
- Response: redesign split protocol, introduce leakage-safe mechanism, re-train
- Added ≈ 2 weeks; resulted in clean, verifiable metrics

Sustainability (Art. 12):

- **Environmental:** ≈ 13 h GPU training, RTX 3060. Inference: 114 ms/page. No cloud compute, no distributed training.
- **Economic:** fully open-source stack (LilyPond, PyTorch, FastAPI). Zero licensing cost.
- **Social:** enables access to music archives. Dataset IP: PRIMuS (freely available); inference on The Real Book does not redistribute copyrighted content.

Objectives achieved

- ✓ **General** — SER 1.23% < 3% target on scan-simulated split
- ✓ **O1** — SoA survey; CRNN+CTC justified against three alternatives
- ✓ **O2** — Synthetic LilyJAZZ dataset from PrIMuS
(46,089 samples; $\approx$ 166k rendered images)
- ✓ **O3** — Scan simulation; clean-to-scanned gap = 0.06 pp
- ✓ **O4** — CRNN+CTC: 1.17% clean / 1.23% scanned
- ✓ **O5** — Error analysis: 90% structural tokens; pitch near-zero
- ✓ **O6** — Full inference pipeline + FastAPI + JSON output
- **O7** — Domain fine-tuning: 36 real staves, qualitative improvement;
no held-out real SER (all 36 staves needed for training)

What was built and demonstrated:

- End-to-end OMR system for jazz lead sheets
- 1.23% aggregate SER on scan-simulated split
- 0.14% melodic SER — pitch is essentially error-free
- 72.7% perfect transcriptions
- Chord recognition: 5.9% CER after fine-tuning (from 69.9% zero-shot)
- 114 ms per page on RTX 3060

Three concrete next steps:

- 1 **Annotate ≥ 200 real staves** → held-out real-domain SER (binding constraint for O7)
- 2 **Tuplet vocabulary** → largest single gap in LMX grammar coverage
- 3 **Learned staff detector** (YOLO/DETR) → robustness on slanted and partial systems

Back to the original problem: digitise *The Real Book* → automatic transposition for B $\flat$, E $\flat$, and F instruments. This system makes that pipeline technically feasible.

CCO1.1 — Algorithmic strategies & performance

CRNN+CTC vs seq2seq vs transformer: justified on compute and data-efficiency grounds (*Ch. 4*).

Greedy vs beam search: < 0.01 pp SER at $8\times$ cost — greedy deployed (*Ch. 6*).

ResNet18 vs VGG: $2\times$ training time, comparable SER.

Deterministic grammar fixer vs learned corrector.

CCO2.4 — Machine learning & large data

CRNN+CTC supervised by CTC loss (*Ch. 4–5*).

$\approx 166,000$ rendered training assets.

Synthetic pre-training then domain fine-tuning applied to both CRNN streams.

Data leakage discovered and remedied (*Ch. 5*).

Both competences declared **in depth** at TFG inscription.

Thank you

Questions?

SER	1.23% scanned / 1.17% clean (4,608 test samples, greedy CTC)
Melodic SER	0.14% — pitch content is essentially error-free
Perfect transcriptions	72.7%
Chord CER	5.9% after fine-tuning (from 69.9% zero-shot)
Latency	114 ms/page on RTX 3060
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